

WALENDRIA PROTOCOL: A TRUSTLESS BURN-OR-RELEASE ESCROW SYSTEM

Introduction

In the world of online buying and selling, both now and in the past, a third party is required as a mediator, a third party is required to hold funds. Like in online marketplaces: the buyer's money is held before being directly forwarded to the seller. The seller then does exactly what the buyer wants, only after the funds are held by the mediator. This transaction system works because one party as mediator is trusted by both sides (buyer and seller). This system is brilliant, because it eliminates the incentive for sellers to easily commit fraud compared to the previous system (direct transfer from buyer to seller).

However, there is one fatal flaw that makes it less than perfect in eliminating friction, namely:

1. The mediator (the fund holder) is a subjective entity. They can be bribed, they can collude with one party.
2. It is cumbersome. We have to search for "a party trusted by both sides" before starting a transaction. The mediator is a subjective entity who may or may not be able to contribute their time and energy to mediate the transaction.

Here, I, as the writer of this whitepaper, will explain a new system of how online transactions should actually work.

Walendria Protocol

This is what I have created: a protocol on the blockchain that eliminates all frictions in online transactions (except eliminating the friction of irrational or idiotic parties, whether buyer or seller). This protocol eliminates the incentives for both seller and buyer to commit fraud. This means the protocol declares that whoever transacts here with malicious intent is already confirmed to be an idiot who does not understand the system.

Let me first discuss the rational and the idiot parties. A rational party can and will profit from acting maliciously (fraud) if "cost < return" (this is basic capitalist science). An irrational party (idiot, stupid, moron) is one who does not care whether "cost < return or not." An example of an irrational person in online transactions is someone who doesn't care if their identity gets viralized, while they gain no equal or greater profit compared to the loss they receive.

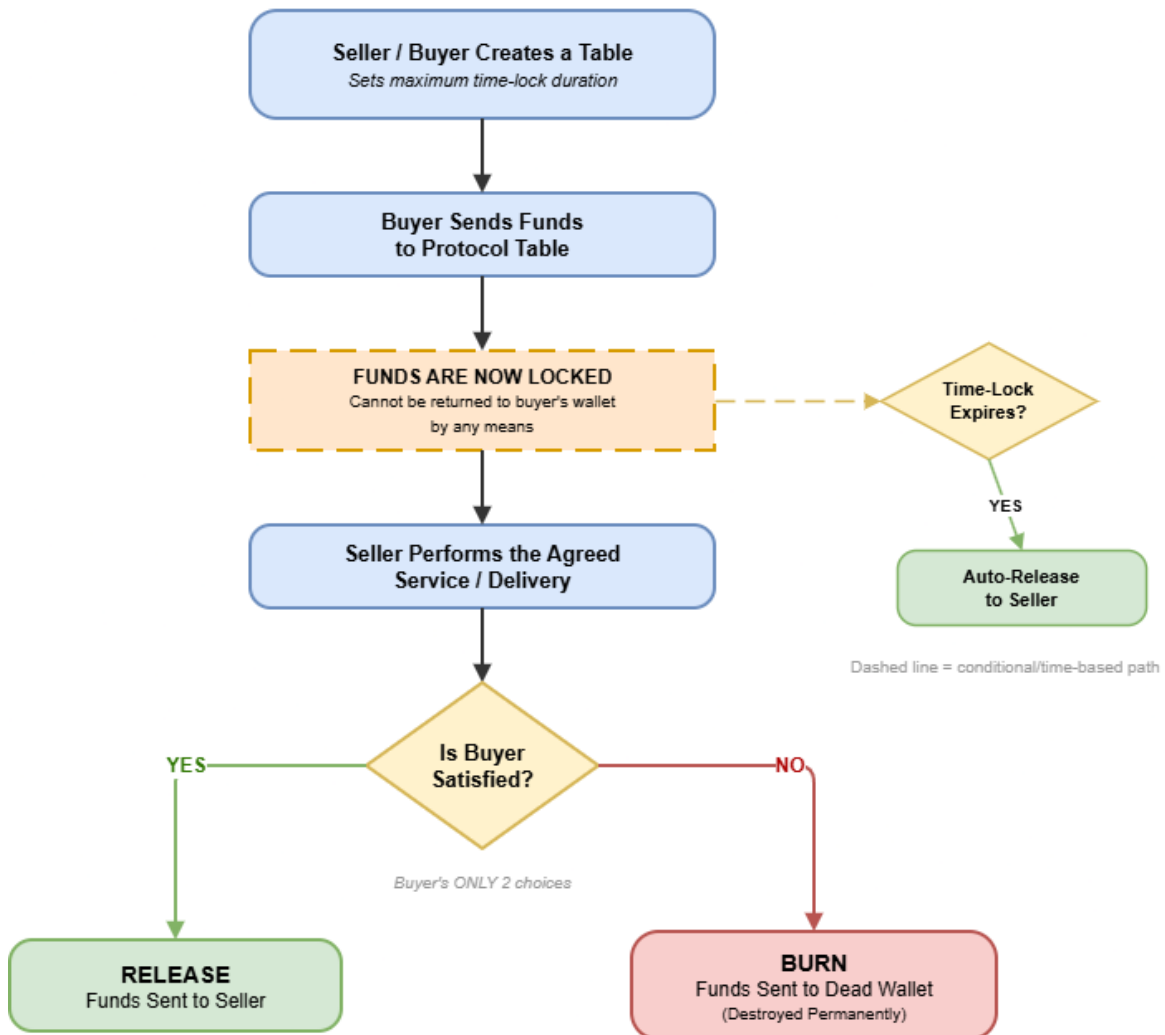
Okay, now that we properly understand the distinction between these two parties, let me explain: this protocol focuses only on objective and rational transactions. Therefore, I designed this protocol entirely on the basis of game theory (Nash equilibrium).

In this protocol, there is no subjective mediator whatsoever. Only calculation. Only mathematics that stays with both parties throughout the transaction. Only a static, objective, and rational program. Yet it remains safe. How can this be?

Here is my protocol design in the language of game theory:

WALENDRIA PROTOCOL FLOWMAP

Trustless Burn-or-Release Escrow System



The seller or buyer creates a table within the protocol, along with a maximum time-lock duration. The buyer sends funds to the protocol table. At this point, these funds cannot be returned to the buyer's wallet in any way. Funds that have been sent to the table can only be released to the seller, or burned (sent entirely to a blockchain dead wallet). The seller sees this, and does what the buyer wants. If the buyer is satisfied, they choose to release the funds to the seller. If they are not satisfied, or feel scammed, or any such reason, they choose to burn (the money cannot come back; it is sent to a dead wallet).

Notice why I only provide two choices for the buyer: "release" or "burn." That is precisely the point that truly eliminates the incentive for the buyer to defraud the seller. But make no mistake. Although this design is very brutal and will make the reader (the buyer) feel strange and think, "I don't want this," look at the other side, at the game theory. Look at the bigger picture: even though the buyer cannot get their money back by any means, look at the situation of the seller. The seller will not receive a single cent from the buyer. They can only receive the money if the buyer

feels satisfied and chooses "release funds to seller." The seller will gain nothing if they intend to act maliciously or commit fraud. Therefore, it can be concluded that this protocol systematically forces the seller to do whatever it takes to make the buyer press the release button so they can get paid.

It is that simple when spoken in game theory. That simple to eliminate the incentive for fraud between both parties. That simple for me to create a law that forces everyone to act "not maliciously," because such behavior is now "unprofitable."

Remaining Frictions

1. This protocol cannot force idiots or people who misclick to act rationally as online transactions should properly and correctly occur. There will always be such parties: those so stupid they do not care about the return:loss ratio, and those so mistaken they click burn even though their intention was release.

Only that.

Further Explanation on the Time-Lock Concept

What is the time-lock auto-release concept? It is an on-chain feature representing the maximum time the escrow exists and remains active (it can actually be set to unlimited, but I will explain later why this feature must exist, in the "To Be Watched Out For" section). Its function is to force the table to be active only within a certain time range agreed upon by both parties. If the escrow's active time runs out, the funds are automatically released to the seller.

To Be Watched Out For

During my thinking about this protocol's design (I share a little story with the reader here), initially, I deployed the contract with the time-lock auto-release concept. But midway, I thought: what if the seller could act maliciously (commit fraud) by stalling until time runs out and the funds are automatically released to them? Well, after that thought reached my brain, I immediately redeployed and removed the time-lock

auto-release concept. However, days later, I realized again that this concept must be brought back into the protocol. I forgot that actually, if a seller intentionally stalls until the escrow time runs out, as long as the escrow is active, the rational buyer will realize this and also realize that what the seller is doing is fraud. Then they can choose to burn (the seller gains no profit from their malicious actions).

I also once thought: what if there is a buyer who threatens the seller in the middle of the active escrow? For example, the buyer says: "Transfer me X amount / give me extra! If not, I will burn," to the seller. That is a classic form of extortion. Even though the buyer will also not profit if their money is actually burned (the seller refuses), it could happen, and the seller might say yes. This sounds like an "infinite money glitch" for the buyer. But let us look at the broader game theory design of this protocol:

A buyer who extorts the seller in exactly this way falls into the category of idiot, stupid, moron (irrational, cannot think). Why? Because "cost > return." When the risk cost is greater than the potential profit, logically, it would be irrational if the action were truly executed. Why can I state that the act of extortion in this protocol is "cost > return"? Let us examine the game theory design again: if the buyer extorts the seller, the seller can choose to refuse, or even report them to others (viralize them). The risk of "getting viralized" for the extorting buyer is real. And if the seller chooses to "refuse," the buyer faces the risk of being "viralized." They suffer a loss and gain no profit. On the other hand, if the seller chooses to "say yes" (then it can be said the seller is an irrational subjective entity) because they do not think about how long the buyer will keep extorting them, then the buyer gets profit. As I mentioned earlier, this protocol cannot prevent irrational entities from continuously acting logically stupid.

In this protocol, I have eliminated the friction of "incentive for rational fraud (cost < return)," but I have not eliminated the friction of "idiot entities transacting." This is already better than the old transaction system where "the mediator is a subjective entity." The "rekber" (escrow) transaction system already works well and smoothly even until now, despite leaving the friction of "non-objective mediator." There is no way this protocol cannot do the same.

Comparison With Other Models

Honestly, this paragraph is more subjective and based on analogies in my mind, but I will convey it anyway. In the offline transaction model, if any party acts maliciously, they will get beaten up or thrown in jail (but there is a probability there, between success or failure). In this protocol's (online) transaction model, if any party acts maliciously, they will get viralized (as a stupid person). But there, too, a probability

remains, between successful viralization or not. Well, that is how it is. The patterns of the world will always repeat.

Why I Do Not Force the Removal of the Remaining Friction

If I were to force the removal of that friction, ideas like "police/judge" would emerge, violating the rule that "this protocol mirrors objective natural law." It is better to leave it be, so as not to add new burdens. I choose to be stoic. The design already mathematically forces all parties to act "non-fraudulently." Everything is finished. There is nothing more I need to think about.

Conclusion

In terms of game theory, this protocol is solid, placing its Nash equilibrium at "the transaction released by the buyer to the seller" (buyer feels satisfied). This protocol is more akin to natural or physical law. It is objective. It does not care about subjective interpretation. Nature sends down rain but remains indifferent to the subjective human who hates the rain now. What can prevent the rain from falling is a person carrying an umbrella. And what can prevent the rain from becoming a flood is planting trees. This logically concludes that if someone hates rain, they must carry an umbrella. And if someone hates floods, they must plant as many trees as possible. The umbrella did not create itself. The umbrella exists because it is a demand of subjectivity upon its interpretation of the objective system (nature that rains).

A Note on the Term "Table"

The author uses the term "table" intentionally. In the Walendria Protocol, the smart contract is deployed only once. Every time someone wants to create an escrow (a transaction instance), the system only creates a "table," not a new contract. Think of it as a room, a table specifically for the buyer and seller within the already-deployed main contract. This design keeps gas fees low. It is an architectural decision, not a typo.